

“best matching partner”



c -transforms

- For a given x , the dual potentials f, g must satisfy $f(x) \leq c(x, y) - g(y) \quad \forall y$
- Since the dual wants to maximize $\int f d\alpha$, want to make $f(x)$ as large as possible
- The largest we can make it before breaking feasibility is $f(x) = \inf_y c(x, y) - g(y)$
- Analogously, best choice for $g(y)$ is $g(y) = \inf_x c(x, y) - f(x)$
- This operation is called c -transform, defined in general as:

$$\forall y : \phi^c(y) \triangleq \inf_x (\phi(x) + c(x, y))$$

- **Intuition:** $\phi^c =$ “**best matching partner**” of ϕ under the transport cost $c(x, y)$

c-conjugacy